

Notes: Kremens, Martin, and Varela (JF 2025)

Long-Horizon Exchange Rate Expectations

Contents

1	Main research questions	3
2	Data and definitions (Section I)	3
2.1	Exchange rates, returns, and the UIP decomposition	3
2.2	Realized objects: IRD, RCA, RXR	4
2.3	Survey objects: SCA and SXR	4
3	Long-horizon predictability: in-sample evidence (Section II.A)	5
3.1	Baseline forecasting regressions (models)	5
3.2	Figure 1: realizations vs survey expectations	5
3.3	Table I: interpretation	5
4	Alternative predictors and the QRP “model” (Section II.B)	7
4.1	A model-free identity and the quanto-implied risk premium	7
4.2	Log-utility benchmark (a tight pricing restriction)	7
4.3	Other proposed predictors (RER, VIX, factor betas, CA/GDP)	7
4.4	Table II: which univariate predictor is best?	7
5	Out-of-sample forecasting (Section II.C)	8
5.1	Forecast errors and out-of-sample R^2	8
5.2	Table III: interpretation	8
6	What informs expectations? (Section III)	9
6.1	Expectation “explanation” regression (model)	9
6.2	Table IV: interpretation (interpretable expectations)	9
7	Is there a “secret sauce”? (Section III.A–B)	9
7.1	In-sample residualization test	9
7.2	Table V: interpretation	10
7.3	Which multivariate predictors maximize R^2 ? (Table VI)	10
7.4	Out-of-sample fitted forecasts vs surveys (Table VII)	10

8 Interpretation: risk-based vs constraint-based views (Section III.C)	11
8.1 A variance-constrained intermediary “model” and its FOCs	11
9 Short versus long horizons (Section IV)	12
9.1 Annualization (definition)	12
9.2 Table VIII: horizon-by-horizon forecasting	12
9.3 Forward expectations decomposition (models)	13
9.4 Table IX: spot vs forward expectations drive predictability	13
9.5 Table X: what informs expectations at shorter horizons?	13
10 Short conclusion	14
11 Conclusion	15
11.1 Summary	15
11.2 Why long-horizon surveys forecast	15
11.3 Practical implication: what the survey adds	16
11.4 Takeaway	16

1 Main research questions

The paper answers three linked questions in a single continuous chain:

1. **Do surveys forecast exchange rates at long horizons?** Yes: two-year (24-month) survey expectations strongly forecast realized currency appreciation and excess returns.
2. **If surveys forecast well, what are they “using”?** A small set of macro-finance variables—QRP, RER, and CA/GDP—spans most of the variation in survey expectations.
3. **Is there extra information in surveys beyond those variables?** No in-sample (and only limited out-of-sample): once you control for the key macro-finance variables, residual survey variation does not forecast.

A separate (puzzling) horizon result closes the loop: **short-horizon survey expectations are weak**, but **long-horizon expectations help forecast short-horizon realizations**. This suggests *horizon-dependent expectation formation*.

2 Data and definitions (Section I)

2.1 Exchange rates, returns, and the UIP decomposition

Let $e_{i,t}$ be the nominal exchange rate in *USD per unit of currency i* . Let $R_{f,t,h}^i$ be the gross h -period risk-free rate in currency i , and $R_{f,t,h}^{\$}$ the gross h -period USD risk-free rate. A dollar investor who converts 1 USD to currency i , invests risk-free, then converts back to USD earns

$$R_{t+h} = R_{f,t,h}^i \frac{e_{i,t+h}}{e_{i,t}}.$$

The paper starts from the standard asset pricing condition for a dollar SDF M_{t+h} :

$$\mathbb{E}_t[M_{t+h}R_{t+h}] = 1, \tag{1}$$

$$\mathbb{E}_t[R_{t+h}] - R_{f,t,h}^{\$} = R_{f,t,h}^{\$} \text{cov}_t(-M_{t+h}, R_{t+h}). \tag{2}$$

Substituting the currency strategy return into (2) yields the key identity:

$$\mathbb{E}_t \left[\frac{e_{i,t+h}}{e_{i,t}} - 1 \right] = \underbrace{\left(\frac{R_{f,t,h}^{\$}}{R_{f,t,h}^i} - 1 \right)}_{\text{UIP term}} + \underbrace{R_{f,t,h}^{\$} \text{cov}_t \left(-M_{t+h}, \frac{e_{i,t+h}}{e_{i,t}} \right)}_{\text{risk premium / residual}}. \tag{3}$$

If the covariance term is ignored, (3) collapses to uncovered interest parity (UIP).

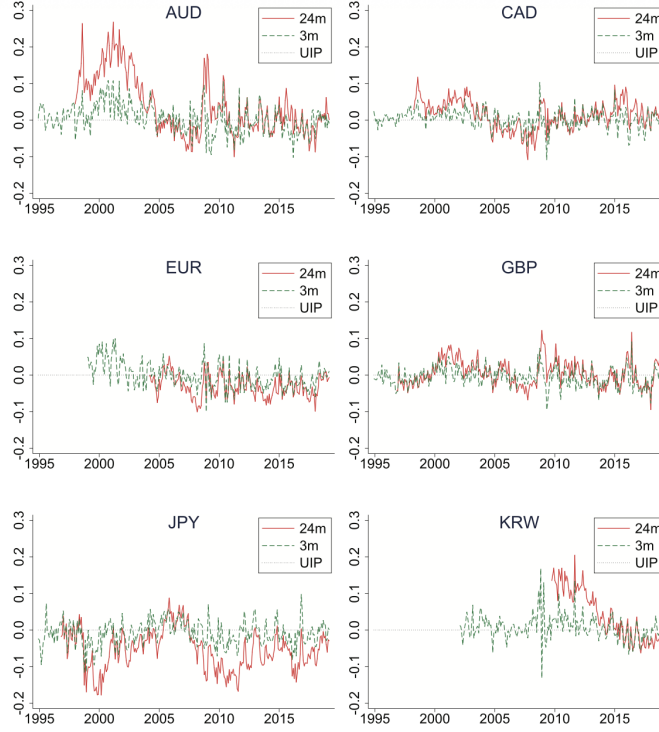


Figure 2. Long-horizon survey expectations over time. This figure plots survey expectations of currency excess returns (SXR , not annualized) at three- and 24-month horizons. UIP predicts that expected excess returns are zero. (Color figure can be viewed at wileyonlinelibrary.com)

2.2 Realized objects: IRD , RCA , RXR

The paper defines:

$$IRD_{i,t,h} \equiv \frac{R_{f,t,h}^{\$}}{R_{f,t,h}^i} - 1, \quad (4)$$

$$RCA_{i,t,h} \equiv \frac{e_{i,t+h}}{e_{i,t}} - 1, \quad (5)$$

$$RXR_{i,t,h} \equiv RCA_{i,t,h} - IRD_{i,t,h}. \quad (6)$$

RXR is a (scaled) currency excess return because it has zero price.

2.3 Survey objects: SCA and SXR

Let $\tilde{\mathbb{E}}_t[\cdot]$ denote the survey consensus expectation (simple average across forecasters). Define:

$$SCA_{i,t,h} \equiv \tilde{\mathbb{E}}_t \left[\frac{e_{i,t+h}}{e_{i,t}} - 1 \right], \quad (7)$$

$$SXR_{i,t,h} \equiv SCA_{i,t,h} - IRD_{i,t,h}. \quad (8)$$

Figure 2 (preview). Plotting SXR over time shows that survey expectations deviate substantially from UIP (UIP would imply $SXR \equiv 0$).

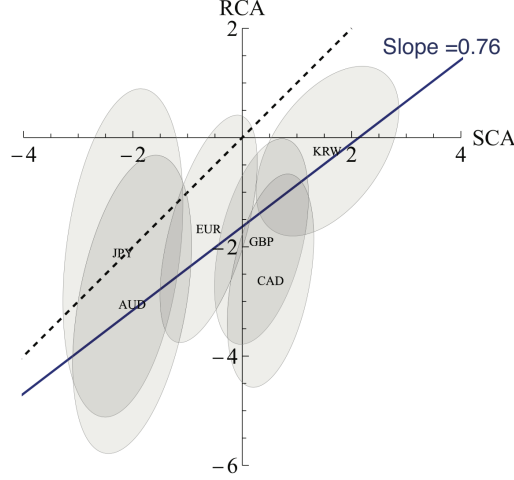


Figure 1. Realized currency appreciation (RCA) versus survey expectations (SCA).

3 Long-horizon predictability: in-sample evidence (Section II.A)

3.1 Baseline forecasting regressions (models)

The core empirical model augments the UIP regression by adding survey expectations:

$$RCA_{i,t,h} = \alpha_h + \gamma_1 SXR_{i,t,h} + \gamma_2 IRD_{i,t,h} + \varepsilon_{i,t,h}. \quad (9)$$

UIP plus “surveys are noise” predicts $\gamma_1 = 0$ and $\gamma_2 = 1$.

The paper also runs the mechanically related regression with realized excess returns on the left-hand side:

$$RXR_{i,t,h} = \alpha_h + \gamma_1 SXR_{i,t,h} + \gamma_2 IRD_{i,t,h} + \varepsilon_{i,t,h}. \quad (10)$$

Because $RXR = RCA - IRD$, the estimated γ_1 is identical in (9) and (10), while γ_2 differs by exactly one.

3.2 Figure 1: realizations vs survey expectations

Figure 1 plots mean realized currency appreciation against mean survey-expected appreciation for each currency, with confidence ellipses summarizing time-series variation. The fitted slope is below but fairly close to the 45-degree line (the paper reports a slope around 0.76), which visually illustrates that survey expectations contain systematic information about realized appreciation.

3.3 Table I: interpretation

Table I reports estimates of (9) and (10) at the 24-month horizon.

Panel A (post-GFC sample, 12/2009–3/2019; $N = 672$).

- **IRD alone forecasts very weakly:** $R^2 \approx 0.03$ for RCA and $R^2 \approx 0.02$ for RXR .

Table I
In-Sample Forecast Performance

This table reports forecasting regressions (9) and (10) of 24-month realized currency appreciation (RCA) and currency excess returns (RXR) on survey-based expectations of currency excess returns (SXR) and interest rate differentials (IRD). The sample is December 2009 to March 2019 (realizations until March 2021) in Panel A and December 1994 to March 2019 in Panel B, and includes AUD, CAD, EUR, GBP, JPY, and KRW against USD. In brackets, we report standard errors obtained from a nonparametric block-bootstrap to account for overlapping observations in long-horizon forecasts.

	RCA				RXR			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Post-GFC Sample (12/2009 to 3/2019)								
SXR		0.726 [0.212]	0.837 [0.251]	0.523 [0.213]		0.726 [0.212]	0.837 [0.251]	0.523 [0.213]
IRD	0.577 [0.599]	1.065 [0.601]	1.147 [0.674]	0.693 [0.548]	-0.423 [0.599]	0.065 [0.601]	0.147 [0.674]	-0.307 [0.548]
Constant (p.a.)	-0.020 [0.012]	-0.017 [0.010]			-0.020 [0.012]	-0.017 [0.010]		
Fixed effects	None	None	Currency	Time	None	None	Currency	Time
R^2	0.031	0.169	0.192	0.564	0.017	0.157	0.180	0.558
Within R^2	0.031	0.169	0.165	0.117	0.017	0.157	0.130	0.174
N	672	672	672	672	672	672	672	672
Panel B. Full Sample (12/1994 to 3/2019)								
SXR		0.865 [0.294]	1.066 [0.269]	0.601 [0.198]		0.865 [0.294]	1.066 [0.269]	0.601 [0.198]
IRD	0.156 [0.575]	0.600 [0.631]	-0.020 [0.707]	0.615 [0.423]	-0.844 [0.575]	-0.400 [0.631]	-1.020 [0.707]	-0.385 [0.423]
Constant (p.a.)	0.004 [0.013]	0.004 [0.013]			0.004 [0.013]	0.004 [0.013]		
Fixed effects	None	None	Currency	Time	None	None	Currency	Time
R^2	0.002	0.145	0.185	0.628	0.058	0.192	0.231	0.649
Within R^2	0.002	0.145	0.173	0.115	0.058	0.192	0.188	0.193
N	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340

- **Adding survey expectations boosts predictability sharply:** in the panel regression, $\hat{\gamma}_1 \approx 0.73$ and is statistically significant, while R^2 rises to about 0.17 for both RCA and RXR .
- With **currency fixed effects**, $\hat{\gamma}_1$ rises to about 0.84 and remains significant, suggesting surveys forecast *within-currency* time-series variation.
- With **time fixed effects**, $\hat{\gamma}_1$ falls to about 0.52 and differs from one, indicating that survey success is not purely cross-sectional; time-series variation is important.

Panel B (full sample, 12/1994–3/2019; $N = 1,340$). The qualitative pattern persists: $\hat{\gamma}_1$ remains close to one in the pooled specification (about 1.07), and surveys continue to deliver materially higher R^2 than IRD alone.

Bottom line. At the two-year horizon, surveys satisfy two success criteria emphasized by the authors: (i) $\gamma_1 > 0$ and statistically significant (qualitative success), and (ii) γ_1 is economically close to 1 (quantitative success).

4 Alternative predictors and the QRP “model” (Section II.B)

4.1 A model-free identity and the quanto-implied risk premium

Kremens and Martin (2019) provide a model-free identity:

$$\mathbb{E}_t \left[\frac{e_{i,t+h}}{e_{i,t}} \right] - \frac{R_{f,t,h}^{\$}}{R_{f,t,h}^i} = \underbrace{\frac{1}{R_{f,t,h}^{\$}} \text{cov}_t^Q \left(\frac{e_{i,t+h}}{e_{i,t}}, R_{t+h} \right)}_{\text{QRP term}} + \underbrace{\text{cov}_t \left(-M_{t+h} R_{t+h}, \frac{e_{i,t+h}}{e_{i,t}} \right)}_{\text{residual}}. \quad (11)$$

Here cov^Q is a *risk-neutral* covariance.

The paper defines the quanto-implied risk premium (QRP) using S&P 500 forwards and quanto forwards:

$$QRP_{i,t,h} \equiv \frac{Q_{i,t,h} - F_t}{R_{f,t,h}^i P_t} = \frac{1}{R_{f,t,h}^{\$}} \text{cov}_t^Q \left(\frac{e_{i,t+h}}{e_{i,t}}, R_{t+h} \right), \quad (12)$$

where F_t is the conventional S&P 500 forward price, $Q_{i,t,h}$ the quanto forward price in currency i , and P_t the spot S&P level.

4.2 Log-utility benchmark (a tight pricing restriction)

In a benchmark where the marginal investor has log utility and is fully invested in the S&P 500, the residual in (11) is zero, implying:

$$\mathbb{E}_t \left[\frac{e_{i,t+h}}{e_{i,t}} \right] - \frac{R_{f,t,h}^{\$}}{R_{f,t,h}^i} = \frac{1}{R_{f,t,h}^{\$}} \text{cov}_t^Q \left(\frac{e_{i,t+h}}{e_{i,t}}, R_{t+h} \right) = QRP_{i,t,h}. \quad (13)$$

Hence QRP is a forward-looking, model-consistent predictor of currency excess returns under a specific benchmark.

4.3 Other proposed predictors (RER, VIX, factor betas, CA/GDP)

The paper compares surveys to:

- **RER:** the real exchange rate (persistent predictor of currency excess returns),
- **VIX:** implied equity volatility as a time-series proxy for risk premia,
- **Dollar and Carry betas** ($\beta^{\$}, \beta^{HML}$) from Lustig–Roussanov–Verdelhan,
- **CA/GDP:** current account balance relative to GDP (related to macro/flow-based views).

4.4 Table II: which univariate predictor is best?

Table II reports univariate R^2 from regressions of 24-month realized RXR on each predictor.

Post-GFC sample. Surveys are best: $R^2(SXR) = 0.157$, beating QRP (0.116) and RER (0.104). VIX and LRV betas deliver moderate R^2 (≈ 0.085), while IRD and CA/GDP are close to zero.

Full sample. QRP is unavailable pre-2009. Surveys remain strongest with $R^2(SXR) = 0.181$; the next best in this sample is VIX (0.105) and RER (0.074).

Interpretation. Surveys compress information from multiple macro-finance sources into a single forecast that outperforms any individual predictor.

5 Out-of-sample forecasting (Section II.C)

5.1 Forecast errors and out-of-sample R^2

Define survey forecast error in currency appreciation:

$$\epsilon_{i,t,t+h}^S \equiv RCA_{i,t,h} - SC A_{i,t,h}.$$

For a competitor model C (random walk RW, or quanto Q), define $\epsilon_{i,t,t+h}^C$ analogously. The Goyal–Welch out-of-sample R^2 is:

$$R_{OS}^2 = 1 - \frac{\sum_i \sum_t (\epsilon_{i,t,t+h}^S)^2}{\sum_i \sum_t (\epsilon_{i,t,t+h}^C)^2}, \quad C \in \{RW, Q\}. \quad (14)$$

A positive R_{OS}^2 means surveys have lower MSE than the competitor.

Because the USD appreciated strongly post-GFC, the paper also uses a dollar-neutral statistic comparing relative forecasts across currencies:

$$\tilde{R}_{OS}^2 = 1 - \frac{\sum_i \sum_j \sum_t (\epsilon_{i,t,t+h}^S - \epsilon_{j,t,t+h}^S)^2}{\sum_i \sum_j \sum_t (\epsilon_{i,t,t+h}^C - \epsilon_{j,t,t+h}^C)^2}. \quad (15)$$

Table II
 R^2 of Alternative Predictors

This table reports the univariate R^2 of regressions of 24-month realized currency excess returns (RXR) on the candidate predictors (and a constant). The last column treats the dollar and carry betas of Lustig, Roussanov, and Verdelhan (2011, 2014) as a single model and reports the bivariate R^2 . The row labeled p -value reports the fraction of bootstrap draws in which the R^2 for the corresponding variable exceeds that for SXR in the full sample (with the exception of QRP which is calculated using the post-GFC sample due to data availability). The post-GFC sample starts in December 2009 and the full sample in December 1994. Predictor variables run through March 2019, with realizations until March 2021.

Sample	Univariate R^2 of RXR on Each Variable								Bivariate (β^{HML} & $\beta^{\$}$)
	SXR	QRP	RER	VIX	β^{HML}	$\beta^{\$}$	IRD	CA / GDP	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post-GFC	0.157	0.116	0.104	0.085	0.072	0.009	0.017	0.000	0.085
Full	0.181		0.074	0.105	0.012	0.003	0.058	0.003	0.016
p -value		[0.191]	[0.069]	[0.193]	[0.030]	[0.048]	[0.097]	[0.010]	[0.198]

Table III
Out-of-Sample Forecast Performance

This table reports out-of-sample R^2 s following Goyal and Welch (2008) for surveys against the random walk (RW) and the Quanto Theory (QRP). The different measures for dollar-based and dollar-neutral returns are defined in equations (14) and (15). The last two lines of the table report p -values for a Diebold-Mariano (DM) test as well as bootstrapped p -values for a test of the null hypothesis that survey expectations and the competitor model perform equally well for all currencies.

Sample	Post-GFC				Full Sample
	RW		QRP		RW
Dollar-Based/Dollar-Neutral	R_{OS}^2	$\tilde{R}_{OS}^2$	R_{OS}^2	$\tilde{R}_{OS}^2$	R_{OS}^2
	(1)	(2)	(3)	(4)	(5)
All	0.1915	0.1499	0.2095	0.0540	0.1341
AUD	0.3125	0.2257	0.2522	0.1268	0.0944
CAD	−0.0054	−0.0639	0.0274	−0.1421	0.0723
EUR	0.3553	0.0711	0.4511	0.0028	0.3726
GBP	0.0841	0.1002	0.1473	−0.0738	0.1964
JPY	0.2024	0.1444	0.1753	0.0395	0.1400
KRW	0.0098	0.4740	0.1604	0.3775	−0.1923
Diebold-Mariano p -value	0.0809	0.0474	0.0278	0.3468	0.1598
Bootstrapped p -value	0.0881	0.0337	0.0382	0.2446	0.0967

5.2 Table III: interpretation

Table III shows that surveys outperform both benchmarks:

- **Versus RW (post-GFC):** $R_{OS}^2 \approx 0.19$ (dollar-based) and $\tilde{R}_{OS}^2 \approx 0.15$ (dollar-neutral).

- **Versus QRP (post-GFC):** $R_{OS}^2 \approx 0.21$.
- **Statistical strength:** outperformance is more robust for dollar-neutral forecasts (bootstrapped $p \approx 0.034$ vs RW), consistent with the idea that the dollar trend is a challenging environment for any dollar-based model.
- **Currency-level heterogeneity:** most currencies show positive out-of-sample gains, except CAD is weaker.

6 What informs expectations? (Section III)

6.1 Expectation “explanation” regression (model)

To understand what forecasters condition on, the paper regresses survey expectations on candidate predictors:

$$SXR_{i,t,h} = \alpha_h + \gamma_1 X_{i,t} + \gamma_2 IRD_{i,t,h} + \varepsilon_{i,t,h}, \quad (16)$$

where $X_{i,t}$ includes subsets of QRP, RER, VIX, CA/GDP, $\beta^\$$, and β^{HML} . Regressors are standardized; coefficients are in percentage points.

6.2 Table IV: interpretation (interpretable expectations)

Univariate columns. QRP and RER are individually highly significant with large R^2 (about 0.40 each). A one-standard-deviation increase in QRP raises survey expected excess returns by about 3.7pp, while a one-standard-deviation increase in RER *lowers* expected excess returns by about 3.1pp.

Multivariate columns. When all predictors are included, QRP and RER remain significant, and CA/GDP becomes significant jointly. Dropping the “irrelevant” predictors barely reduces R^2 (about 0.536 to 0.528), so **QRP + RER + CA/GDP** span essentially all explainable variation in long-horizon survey expectations.

Interpretation. Survey expectations are not a black box: they look like a disciplined mapping from a small set of macro-finance state variables into conditional expected returns.

7 Is there a “secret sauce”? (Section III.A–B)

7.1 In-sample residualization test

Define $\widehat{SXR}$ and $\varepsilon(SXR)$ as the fitted value and residual from projecting SXR on $\{QRP, RER, CA/GDP\}$. If residual survey components contain proprietary information, $\varepsilon(SXR)$ should forecast realizations.

Table IV
What Informs Exchange Rate Expectations?

This table presents regressions of 24-month survey expectations of currency excess returns (SXR, not annualized) onto a constant and various standardized financial and macroeconomic variables: the interest rate differential (IRD), the quanto-implied risk premium (QRP), the real exchange rate (RER), the 30-day S&P implied volatility index (VIX), the current account-to-GDP ratio (CA/GDP), and the 24-month rolling monthly beta of the exchange rate on the dollar and carry factors of Lustig, Roussanov, and Verdelhan (2011, 2014), respectively ($\beta^{\$}$, β^{HML}). Coefficients are expressed in percentage points. Standard errors in parentheses are clustered at the currency and time level. We report asterisks indicating significance at 10%, 5%, and 1%, respectively, for convenience given the large number of columns and regressors.

	Survey Excess Returns (SXR)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
IRD	-2.041 (1.142)	0.851 (0.612)	-3.467 ** (0.905)	-1.895 (1.192)	-1.913 (1.460)	-1.950 (1.232)	0.283 (0.861)	
QRP		3.737 *** (0.491)					3.056 *** (0.239)	3.052 *** (0.515)
RER			-3.090 ** (0.927)				-1.763 ** (0.678)	-1.807 ** (0.598)
VIX				0.732 (0.798)			0.141 (0.304)	
CA / GDP				-0.297 (1.406)		-1.274 ** (0.386)	-1.287 *** (0.300)	
$\beta^{\$}$						-0.156 (0.885)	-0.308 (0.807)	
β^{HML}						1.058 *** (0.244)	0.347 (0.271)	
R^2	0.138	0.402	0.387	0.155	0.140	0.175	0.536	0.528
N	672	639	672	672	672	672	639	639

Table V
Do Survey Respondents Have a Secret Sauce?

This table reports forecasting regressions of 24-month realized currency appreciation (RCA) onto a constant, interest rate differentials (IRD), survey-based excess returns (SXR), and the fitted values ($\widehat{SXR}$) and residuals ($\varepsilon(SXR)$) of SXR. Fitted values and residuals are obtained from the trivariate specification in column (8) of Table IV. The sample runs from December 2009 to March 2019 (realizations until March 2021) and includes AUD, CAD, EUR, GBP, JPY, and KRW against USD. In brackets, we report standard errors obtained from a nonparametric block-bootstrap to account for overlapping observations in long-horizon forecasts.

	RCA		
	(1)	(2)	(3)
IRD	1.137 [0.747]	1.563 [0.855]	1.559 [0.900]
SXR	0.740 [0.246]		
$\widehat{SXR}$		1.415 [0.841]	1.414 [0.832]
$\varepsilon(SXR)$			0.177 [0.232]
R^2	0.170	0.252	0.256
N	639	639	639

7.2 Table V: interpretation

Table V regresses 24-month realized RCA on IRD plus either SXR , $\widehat{SXR}$, and/or $\varepsilon(SXR)$:

- Using SXR : coefficient ≈ 0.74 with $R^2 \approx 0.17$.
- Using $\widehat{SXR}$ alone: coefficient rises (about 1.42) and R^2 increases to about 0.25.
- Adding the residual: its coefficient is small (≈ 0.18) and adds almost nothing to R^2 (to about 0.256).

Thus, **there is no in-sample secret sauce**: the predictive content of surveys is essentially the content of the macro-finance variables.

7.3 Which multivariate predictors maximize R^2 ? (Table VI)

Table VI reports the R^2 -maximizing predictor sets for realized 24-month RXR :

- Best univariate: SXR with $R^2 = 0.157$.
- Best bivariate: (QRP, RER) with $R^2 = 0.260$.
- Best trivariate: (QRP, RER, VIX) with $R^2 = 0.314$.
- Adding all variables yields $R^2 \approx 0.359$; excluding SXR barely changes R^2 (≈ 0.357), implying surveys add little incremental power once the full predictor set is present.

7.4 Out-of-sample fitted forecasts vs surveys (Table VII)

Table VII benchmarks surveys against an econometrician who forecasts returns out-of-sample using an expanding window and the macro-finance predictors (excluding QRP pre-2009). Key interpretation:

Table VI
 R^2 -Maximizing Predictors

This table reports the R^2 -maximizing univariate, bivariate, etc., specifications in regressions of 24-month realized currency excess returns (RXR) onto a constant and combinations of various standardized candidate predictors. The last column reports the specification with all variables except SXR. Coefficients are expressed in percentage points.

Coefficient Estimates in R^2 -Maximizing Specifications					
	Univariate (1)	Bivariate (2)	Trivariate (3)	8-Variate (4)	Excl. SXR (5)
SXR	3.916	.	.	0.639	.
QRP	.	3.705	3.055	0.988	1.343
RER	.	-3.715	-4.183	-4.805	-5.010
VIX	.	.	2.459	2.920	2.936
IRD	.	.	.	-3.018	-2.985
CA/GDP	.	.	.	0.528	0.379
β^8	.	.	.	1.170	1.135
β^{HML}	.	.	.	1.827	1.867
R^2	0.157	0.260	0.314	0.359	0.357

Table VII
Surveys versus Out-of-Sample Fitted Forecasts

This table reports forecasting regressions of 24-month realized currency excess returns (RXR) on a constant, survey-based expectations of excess returns (SXR), out-of-sample forecasts of RXR ($\widehat{RXR}^{OOS}$) and SXR ($\widehat{SXR}^{OOS}$), and the residuals of the latter ($\varepsilon(SXR)^{OOS}$). Fitted values and residuals are obtained from expanding-window regressions of RXR and SXR, respectively, on IRD, RER, VIX, CA/GDP, β^{HML} , and β^8 . The estimation window is initialized using dependent variables observed until November 2009 and then expands one month at a time. The sample for the forecasting regressions reported in this table runs from December 2009 to March 2019 (realizations until March 2021). In brackets, we report standard errors obtained from a nonparametric block-bootstrap.

RXR					
	(1)	(2)	(3)	(4)	(5)
SXR	0.713 [0.200]		0.559 [0.218]		
$\widehat{RXR}^{OOS}$		0.745 [0.363]	0.454 [0.474]		
$\widehat{SXR}^{OOS}$				1.285 [0.696]	1.407 [0.763]
$\varepsilon(SXR)^{OOS}$					0.540 [0.206]
R^2	0.157	0.112	0.191	0.131	0.207
N	672	672	672	672	672

- The econometrician forecast $\widehat{RXR}^{OOS}$ predicts, but with lower R^2 (≈ 0.112) than surveys (≈ 0.157).
- Combining both raises R^2 only modestly (≈ 0.191), and the econometrician forecast is largely driven out.
- Projecting surveys out-of-sample onto the predictors yields $\widehat{SXR}^{OOS}$ with $R^2 \approx 0.131$; the residual then predicts strongly. The paper shows this residual mainly reflects missing QRP information, not true proprietary forecasting skill.

8 Interpretation: risk-based vs constraint-based views (Section III.C)

The paper emphasizes that the same empirical patterns can be consistent with:

- Risk-based pricing** (a stochastic discount factor with risk aversion), and/or
- Intermediary constraints** (risk-neutral intermediaries with balance-sheet/variance constraints).

8.1 A variance-constrained intermediary “model” and its FOCs

Consider a risk-neutral intermediary who chooses portfolio weights $w_1, \dots, w_N$ on excess returns $R_{1,t+h}^e, \dots, R_{N,t+h}^e$ to maximize expected return subject to a risk-neutral variance constraint:

$$\max_{w_1, \dots, w_N} \mathbb{E}_t \left[\sum_k w_k R_{k,t+h}^e \right] \quad \text{s.t.} \quad \text{var}_t^Q \left(\sum_k w_k R_{k,t+h}^e \right) \leq \kappa.$$

The first-order conditions imply:

$$\mathbb{E}_t[R_{i,t+h}^e] = \lambda \text{cov}_t^Q(R_{i,t+h}^e, R_{t+h}),$$

where λ is the Lagrange multiplier on the variance constraint and R_{t+h} is the intermediary's portfolio return.

Applying this to the currency excess return yields:

$$\mathbb{E}_t \left[\frac{e_{i,t+h}}{e_{i,t}} \right] - \frac{R_{f,t,h}^S}{R_{f,t,h}^i} = \lambda \text{cov}_t^Q \left(\frac{e_{i,t+h}}{e_{i,t}}, R_{t+h} \right). \quad (17)$$

If R_{t+h} is (or is closely correlated with) the S&P 500 return, the right-hand side is proportional to QRP. Hence QRP can arise from risk-based logic *or* constraint-based logic: the data alone do not uniquely identify the mechanism.

9 Short versus long horizons (Section IV)

9.1 Annualization (definition)

To compare horizons, the paper annualizes variables:

$$SCA_{i,t,h}^* \equiv \frac{12}{h} SCA_{i,t,h}, \quad (18)$$

and analogously for SXR^* , RXR^* , and IRD^* .

9.2 Table VIII: horizon-by-horizon forecasting

Short-horizon forecasts are weak. Columns (1)–(3) show that one-, three-, and 12-month survey expectations have coefficients close to zero and are not statistically different from zero. Only at 24 months does the coefficient move toward one and become strongly significant (replicating Table I).

Long-horizon expectations predict short-horizon realizations. Columns (5)–(7) show that SXR_{24}^* predicts RXR_1^* , RXR_3^* , and RXR_{12}^* with coefficients near one (especially in the full sample), suggesting that the component of expected excess returns embedded in long-run forecasts materializes fairly smoothly over the 24-month window.

But short-horizon expectations do not predict long-horizon realizations. Columns (8)–(10) show that SXR_1^* and SXR_3^* are close to useless predictors of RXR_{24}^* .

Interpretation. The puzzle is not that short-horizon *realizations* are unpredictable—they are predictable using long-horizon expectations. Rather, the puzzle is that **short-horizon expectations are noisy**: forecasters appear to use a different model at short horizons.

9.3 Forward expectations decomposition (models)

Define (in logs) the forward expectation between horizons h and $H > h$:

$$sca_{i,t}^{h,H} \equiv \log\left(\frac{1 + SCA_{i,t,H}}{1 + SCA_{i,t,h}}\right) = sca_{i,t,H} - sca_{i,t,h}, \quad (19)$$

where $sca_{i,t,h} \equiv \log(1 + SCA_{i,t,h})$ and $rca_{i,t,h} \equiv \log(1 + RCA_{i,t,h})$. This yields the additive decomposition:

$$sca_{i,t,24} = sca_{i,t,3} + sca_{i,t}^{3,12} + sca_{i,t}^{12,24}. \quad (20)$$

Table VIII
Forecast Performance across Horizons

This table reports forecasting regressions of annualized one-, three-, 12-, and 24-month **realized currency excess returns (RXR^*)** on a constant and survey-based expectations of annualized excess returns (SXR^*) and interest rate differentials (IRD^*). The sample is an unbalanced panel and includes AUD, CAD, EUR, GBP, JPY, and KRW against USD. The horizon for IRD^* is always equal to that of RXR^* . In brackets, we report standard errors, clustered by currency and time for one-month realizations and obtained from a nonparametric block-bootstrap to account for overlapping observations in three-, 12-, and 24-month forecasts.

	Forecasting RXR_h^* with SXR_h^*				RXR_h^* with SXR_{24}^*				RXR_{24}^* with SXR_h^*			
	1M (1)	3M (2)	12M (3)	24M (4)	1M (5)	3M (6)	12M (7)	1M (8)	3M (9)	12M (10)		
Panel A. Post-GFC Sample (12/2009 to 3/2019)												
SXR^*	0.088 [0.067]	0.093 [0.102]	0.237 [0.215]	0.726 [0.212]	1.548 [0.857]	0.911 [0.543]	0.683 [0.384]	0.007 [0.009]	0.037 [0.031]	0.282 [0.130]		
IRD^*	-1.112 [0.856]	-1.066 [0.998]	-0.689 [0.890]	0.065 [0.601]	-0.083 [0.763]	-0.498 [1.057]	-0.406 [1.056]	-0.425 [0.613]	-0.405 [0.612]	-0.221 [0.642]		
R^2	0.011	0.014	0.038	0.157	0.018	0.028	0.070	0.019	0.025	0.071		
N	672	672	672	672	672	672	672	672	672	672		
Panel B. Full Sample (12/1994 to 3/2019)												
SXR^*	0.013 [0.041]	-0.010 [0.081]	0.380 [0.237]	0.865 [0.294]	1.007 [0.356]	0.879 [0.411]	0.839 [0.379]	0.012 [0.010]	0.071 [0.039]	0.447 [0.173]		
IRD^*	-1.233 [0.467]	-1.037 [0.738]	-0.835 [0.717]	-0.400 [0.631]	-0.758 [0.503]	-0.610 [0.807]	-0.611 [0.732]	-0.850 [0.631]	-0.816 [0.572]	-0.623 [0.563]		
R^2	0.006	0.011	0.073	0.192	0.013	0.029	0.105	0.061	0.075	0.145		
N	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340		

Table IX

In-Sample Predictability: Spot and Forward Expectations

This table presents regressions of realized currency appreciation onto a constant and spot and forward survey expectations of currency appreciation. Standard errors in brackets are obtained from a nonparametric block-bootstrap to account for overlapping observations in realizations. Observations range from December 1994 to March 2019.

	RCA_{24}	rca_{24}			rca_3
	(1)	(2)	(3)	(4)	(5)
SCA_{24}	0.812 [0.269]				
sca_{24}		0.804 [0.266]			
sca_3			0.252 [0.192]	0.246 [0.226]	-0.062 [0.086]
$sca^{3,12}$			1.070 [0.943]		
$sca^{12,24}$			1.136 [0.897]		
$sca^{3,24}$				1.102 [0.361]	0.188 [0.087]
R^2	0.139	0.138	0.162	0.162	0.030
N	1,340	1,340	1,340	1,340	1,340

9.4 Table IX: spot vs forward expectations drive predictability

Table IX uses this decomposition to show:

- Predictability is similar in levels and logs (columns (1)–(2)).
- The **three-month spot expectation** has a coefficient economically close to zero in predicting 24-month realizations.
- The **forward expectations** (beyond the short horizon) carry the predictive content: the long-horizon forward component has a coefficient near one and is significant.
- Even more starkly, forward expectations from 3 to 24 months predict *three-month* realizations (column (5)), while three-month spot expectations do not.

9.5 Table X: what informs expectations at shorter horizons?

Table X repeats the “what informs expectations” regression across horizons:

- At 1–3 months, the macro-finance variables explain little ($R^2 \approx 0.03$ to 0.07); only the dollar beta is statistically significant.

Table X
What Informs Exchange Rate Expectations at Shorter Horizons?

This table presents regressions of survey expectations of currency excess returns (SXR, not annualized) at one-, three-, 12-, and 24-month horizons on a constant and standardized financial and macroeconomic variables: the horizon-matched interest rate differential (IRD), the 24-month quanto-implied risk premium (QRP), the real exchange rate (RER), the 30-day S&P implied volatility index (VIX), the current account balance relative to GDP (CA/GDP), and the 24-month rolling monthly beta of the exchange rate on the dollar and carry factors of Lustig, Roussanov, and Verdelhan (2011, 2014), respectively (β^S , β^{HML}). Coefficients are expressed in percentage points. Standard errors in parentheses are clustered at the currency and time level. *, **, and *** indicate significance at 10%, 5%, and 1%, respectively. Observations range from December 2009 to March 2019.

	Survey Excess Returns (SXR)			
	Horizon (Months)			
	1 (1)	3 (2)	12 (3)	24 (4)
IRD	-0.114 (0.274)	-0.235 (0.420)	0.871 (0.728)	0.283 (0.861)
QRP	0.094 (0.218)	0.436 (0.231)	2.277 *** (0.173)	3.056 *** (0.239)
RER	-0.098 (0.139)	-0.128 (0.248)	-0.504 (0.549)	-1.763 ** (0.678)
VIX	0.376 (0.305)	0.423 (0.300)	0.519 (0.300)	0.141 (0.304)
CA/GDP	-0.049 (0.102)	-0.080 (0.113)	-0.716 ** (0.201)	-1.274 ** (0.386)
β^S	0.387* (0.191)	0.665 ** (0.225)	0.012 (0.535)	-0.308 (0.807)
β^{HML}	-0.065 (0.269)	-0.066 (0.253)	-0.218 (0.222)	0.347 (0.271)
R^2	0.030	0.070	0.311	0.536
N	639	639	639	639

- At 12 months, R^2 jumps sharply (≈ 0.31), and QRP and CA/GDP become strongly significant.
- At 24 months, QRP, RER, and CA/GDP are the key drivers and R^2 is very high (≈ 0.54), consistent with Table IV.

This reinforces the horizon-dependent model formation view: **short-run expectations do not seem to condition on the variables that forecast returns at long horizons.**

10 Short conclusion

The paper delivers three takeaways that fit together tightly:

- Long-horizon surveys forecast currency appreciation and excess returns, both in and out of sample.
- Survey expectations are interpretable and largely spanned by QRP, RER, and CA/GDP.
- Once those variables are controlled for, residual survey variation has essentially no predictive power for returns (no “secret sauce”).

The remaining open issue is the horizon puzzle: forecasters’ short-run expectations appear noisy and weakly related to macro-finance predictors, even though long-run expectations predict short-run realizations.

11 Conclusion

11.1 Summary

This paper concludes that **long-horizon exchange-rate expectations reported in professional surveys contain strong, economically meaningful predictive information**, and that this predictive content is largely **interpretable through a small set of public macro-finance state variables**.

The key conclusions form one chain:

- (i) **Long-horizon survey expectations forecast exchange rates.** Two-year (24-month) survey expectations forecast realized currency appreciation and currency excess returns with substantial explanatory power, and they outperform benchmark predictors such as the interest-rate differential (UIP), as well as many standard macro/finance predictors.
- (ii) **Surveys forecast by embedding time-varying expected excess returns (risk premia).** Survey-implied expected excess returns vary strongly over time and across currencies, indicating that forecasters do not treat UIP as a good approximation at long horizons. Instead, survey expectations behave as if forecasters are tracking a time-varying risk premium component.
- (iii) **Survey expectations are largely spanned by three interpretable predictors.** Most of the variation in long-horizon survey expected excess returns can be explained by a small set of variables: the **quanto-implied risk premium (QRP)**, the **real exchange rate (RER)**, and the **current account-to-GDP ratio (CA/GDP)**. These variables jointly account for a large share of the level and time variation in survey expectations.
- (iv) **There is little “secret sauce” beyond those variables.** After projecting survey expectations onto (QRP, RER, CA/GDP), the residual component of survey expectations has essentially no additional in-sample forecasting power for realized currency movements. In other words, surveys forecast well mainly because they aggregate and filter information already present in observable macro-finance variables.
- (v) **Horizon dependence is central: short-horizon expectations are weak, but long-horizon expectations predict short-horizon realizations.** Short-horizon survey expectations (e.g., 1–3 months) exhibit limited forecasting power. Yet long-horizon survey expectations forecast not only long-horizon outcomes but also a meaningful portion of short-horizon realized exchange-rate changes. This implies that the expected-return component embedded in long-horizon forecasts is persistent and materializes gradually over time, while short-horizon expectations appear comparatively noisy.

11.2 Why long-horizon surveys forecast

The paper’s intuition can be summarized as **“persistent risk premia + horizon-specific expectation formation.”**

1) Exchange-rate movements contain a risk-premium component that is persistent. At long horizons, realized exchange-rate changes reflect not only interest-rate differentials (UIP) but also a time-varying compensation for risk (or for intermediary constraints). If this expected excess return component is persistent, then a forecast formed today about the *two-year* expected excess return will also have predictive content for the *next few months* of realized returns.

2) QRP is a forward-looking proxy for currency risk premia. The QRP is extracted from quanto markets and captures a risk-neutral covariance object tied to the comovement between the currency payoff and equity-market payoffs. If the pricing kernel (or marginal intermediary constraint) loads on equity risk, then QRP is informative about currency expected excess returns. Survey expectations co-move strongly with QRP, suggesting that forecasters implicitly condition on information that is closely related to derivative-implied risk premia.

3) RER and CA/GDP summarize slow-moving macro forces. The real exchange rate captures valuation or “misalignment” relative to a long-run anchor. The current account captures external imbalances and medium-run adjustment pressures. Both are slow-moving and naturally suited to long-horizon forecasting, and surveys appear to incorporate them.

4) Why short-horizon survey expectations are weak. Short-horizon exchange-rate changes are dominated by high-frequency noise, rapid news flow, and microstructure effects, so professional forecasters may not (or cannot) map the same macro-finance state variables into precise short-horizon predictions. Instead, they form a more reliable *long-horizon* view (a persistent expected-return signal), which then shows up in short-horizon realizations because the risk premium is realized gradually.

11.3 Practical implication: what the survey adds

The paper implies that surveys are valuable less because they contain private information, and more because they serve as an **aggregation device**: they compress multiple macro-finance signals into a single expectation measure with strong forecasting performance. This makes long-horizon surveys a useful empirical proxy for **expected currency excess returns** in applications where the researcher needs an observable expectation object.

11.4 Takeaway

Professional two-year exchange-rate expectations forecast currency appreciation and excess returns because they largely reflect a persistent risk-premium signal that is spanned by QRP, the real exchange rate, and current-account imbalances; after controlling for these variables, residual survey variation adds little extra forecasting power, and the remaining puzzle is the strong horizon dependence of expectation formation.